MINI PROJECT REPORT

■ CAMPUS LOST & FOUND AI

AI-Powered Web Application for Intelligent Campus Lost-and-Found Management

Submitted by
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Course
B.Tech – Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Project Type
Mini Project

Development Platform
Base44 AI

Live Application
https://flat-campus-find-sync.base44.app

CERTIFICATE

This is to certify that Avula Nandini, a student of B.Tech – Computer Science and Engineering (Artificial Intelligence & Machine Learning), has successfully completed the mini project titled “Campus Lost & Found AI” as part of the academic requirements for the mini project. The project was developed as an AI-assisted web application using Base44 AI.

Project Guide	Head of Department	Student
_____	_____	_____
Signature & Date	Signature & Date	Signature & Date

DECLARATION

I, Avula Nandini, hereby declare that the mini project titled “Campus Lost & Found AI” is my academic project work. The application was developed using Base44 AI and the concepts, workflow, database structure, and AI-assisted matching approach described in this report. The live application is available at https://flat-campus-find-sync.base44.app.

Place: _____

Date: _____

Signature: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to my institution, faculty members, project guide, and everyone who provided guidance and support during the development of this mini project. I also acknowledge the role of modern AI-assisted development tools in helping transform the project idea into a functional web application. Their support and encouragement helped me understand practical application development, user interface design, database-driven workflows, and the use of AI-assisted matching in a real-world problem.

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1. ABSTRACT

Campus Lost & Found AI is an AI-powered web application designed to simplify the process of reporting, searching, matching, and recovering lost items within a college campus. Students, faculty, and staff frequently lose belongings such as ID cards, books, bags, wallets, calculators, headphones, water bottles, keys, and other accessories.

The proposed system provides a centralized digital platform where users can report lost or found items by entering item details and uploading images. The application uses AI-assisted matching logic to compare available item information, including image characteristics, category, color, description, location, and date, and then presents potential matches with a match score.

The application was developed with Base44 AI, which was used to create the web interface, application workflows, data management, authentication, and AI-assisted functionality. The live application is available at https://flat-campus-find-sync.base44.app.

2. INTRODUCTION

A college campus is a dynamic environment in which many personal belongings are carried between classrooms, laboratories, libraries, canteens, hostels, transport areas, and other common spaces. When an item is lost, the owner may not know whether another person has found it. Similarly, a person who finds an item may not know how to locate its owner.

A digital lost-and-found platform can connect these two sides. Campus Lost & Found AI extends the conventional idea by adding intelligent matching. Instead of requiring users to manually search every report, the system can rank potentially related reports and make the recovery process more efficient.

3. PROBLEM STATEMENT

Students and staff often lose personal belongings within college campuses, while found items may remain unclaimed because there is no efficient centralized system to connect the person who lost an item with the person who found it. Traditional lost-and-found processes rely on manual searching, notices, and communication. Therefore, there is a need for an intelligent web application that allows users to report lost and found items and identifies potentially similar reports using image and item information.

4. OBJECTIVES

Provide a centralized platform for campus lost-and-found management.

Allow students and staff to report lost and found items digitally.

Support item image uploads and structured item descriptions.

Use AI-assisted matching to identify potentially similar reports.

Generate a match score to rank possible matches.

Provide search, filtering, notifications, and item-status tracking.

Provide an administrative dashboard for monitoring reports and matches.

5. SCOPE OF THE PROJECT

The project is intended for educational institutions and can initially be deployed for a single college campus. The scope includes user authentication, lost and found reporting, image upload, item search, AI-assisted matching, notifications, dashboards, and administrative management. The system can later be expanded to multiple campuses, mobile applications, advanced computer vision, multilingual support, and campus-map integration.

6. EXISTING SYSTEM

In a conventional campus environment, lost-and-found activities are often managed through physical notice boards, social-media groups, class groups, office registers, or verbal communication. These approaches scatter reports across channels, make manual searching time-consuming, provide no systematic image comparison, and make old announcements difficult to locate.

7. PROPOSED SYSTEM

The proposed system provides one web-based platform for reporting, searching, matching, and managing lost and found items. Users can submit item details and images, while the system maintains structured records. The AI-assisted matching process evaluates multiple signals and ranks possible matches. This creates a more organized and scalable workflow than manual lost-and-found management.

8. SYSTEM REQUIREMENTS

8.1 Hardware Requirements

- Laptop or desktop computer with a modern web browser.
- Minimum 4 GB RAM recommended for development and testing.
- Stable internet connection.

8.2 Software Requirements

- Windows, Linux, or macOS.
- Modern browser such as Chrome, Edge, or Firefox.
- Base44 AI.
- Internet connection.
- GitHub, if source/documentation hosting is required.

9. SYSTEM ARCHITECTURE

The application follows a web-based architecture in which users interact with a responsive interface, application workflows process submitted information, database entities store reports, and the AI-assisted matching layer evaluates possible relationships between lost and found items.

Layer	Responsibility
Presentation	Responsive pages, forms, dashboards and navigation.
Authentication	Registration, login, roles and profiles.
Application	Reports, search, filters, statuses and notifications.
Data	Users, lost items, found items, matches and notifications.

Layer	Responsibility
AI-Assisted Matching	Image/item comparison, similarity signals, ranking and match score.

10. MODULES

10.1 User Authentication

Users can register, log in, log out, and manage their profile. Role-based access can distinguish normal users from administrators.

10.2 Report Lost Item

Users submit item name, category, description, image, color, brand, date, time, location, and other identifying details.

10.3 Report Found Item

Users submit information about items they have found, including an image and the location/date where the item was found.

10.4 AI-Assisted Matching

The system compares lost and found reports using available image and textual attributes and ranks possible matches.

10.5 Search and Filtering

Users can search by keyword and filter reports by category, color, location, date, and status.

10.6 Match Results

Potential matches are shown with images, item details, match score, and relevant evidence.

10.7 Notifications

Users can be notified about potential matches and important report-status changes.

10.8 User Dashboard

Users can view their lost items, found items, possible matches, and returned items.

10.9 Admin Dashboard

Administrators can monitor reports, users, matches, statuses, and basic platform analytics.

11. AI/ML COMPONENT

Artificial Intelligence is the central intelligent component of the project. The application is designed to compare multiple attributes rather than relying on a single field. The matching process can consider image characteristics, category, color, textual description, location relevance, and date.

Matching Signal	Purpose
Image Similarity	Checks whether two uploaded item images appear visually related.
Category Similarity	Checks whether both reports describe the same type of item.

Matching Signal	Purpose
Color Similarity	Uses reported color as an additional matching signal.
Description Similarity	Compares descriptive characteristics and identifying details.
Location Relevance	Considers whether the lost and found locations are related.
Date Relevance	Considers whether the reports occurred within a reasonable time window.

A combined match score can rank candidates. Example: image similarity 90%, category similarity 100%, color similarity 95%, description similarity 85%, and overall match score 91%. Actual production accuracy depends on the connected AI/computer-vision implementation and the quality of submitted data.

12. DATABASE DESIGN

12.1 Users

Field	Description
User ID	Unique user identifier
Name	User name
Email	User email
Role	Student/Staff/Admin
Created Date	Account creation date

12.2 Lost Items

Field	Description
Item ID	Unique item ID
User ID	Reporting user
Item Name	Name of item
Category	Item category
Description	Item description
Image	Uploaded image
Color	Item color
Location	Lost location
Date	Lost date
Status	Current status

12.3 Found Items

Field	Description
Item ID	Unique item ID
User ID	Reporting user

Field	Description
Item Name	Name of item
Category	Item category
Description	Item description
Image	Uploaded image
Color	Item color
Location	Found location
Date	Found date
Status	Current status

12.4 Matches

Field	Description
Match ID	Unique match identifier
Lost Item ID	Related lost item
Found Item ID	Related found item
Match Score	Similarity percentage/score
Status	Pending/Confirmed/Rejected
Created Date	Match creation date

13. APPLICATION WORKFLOW

The workflow begins when a user registers or logs in. The user reports a lost or found item, uploads an image, and enters structured information. The application stores the report and initiates matching. Existing reports are evaluated and possible matches are ranked. The user reviews a candidate, confirms or rejects it, and the item can eventually be marked as returned.

Step	Process
1	User registers or logs in.
2	User selects Report Lost or Report Found.
3	User enters item information and uploads an image.
4	Application stores the report.
5	AI-assisted matching compares existing reports.
6	Potential matches are ranked by match score.
7	User reviews a possible match.
8	Match is confirmed or rejected.
9	Item status is updated to Returned/Closed.

14. USER INTERFACE AND FEATURES

- Landing page with project purpose and action buttons.
- Lost-item reporting form with image upload.
- Found-item reporting form with image upload.
- Search page with keyword and filters.
- Match results with candidate cards and scores.
- Personal dashboard for reports and statuses.
- Notifications for potential matches and updates.
- Admin dashboard for report management and analytics.

15. TESTING

Testing verifies that the major workflows behave as expected. The following functional test cases cover the core application.

Test Case	Expected Result	Status
User registration	Account is created successfully	Pass
User login	Authenticated dashboard is displayed	Pass
Lost item submission	Lost report is stored	Pass
Found item submission	Found report is stored	Pass
Image upload	Image is accepted and displayed	Pass
Search and filtering	Relevant reports are displayed	Pass
Potential matching	Possible matches are displayed	Pass
Match status update	Status changes correctly	Pass
Notification workflow	Relevant update is shown	Pass
Admin access	Authorized admin can access dashboard	Pass

16. ADVANTAGES

- Centralized lost-and-found management.
- Reduces manual searching and repeated announcements.
- Image-supported reporting improves item identification.
- AI-assisted matching prioritizes potentially related reports.
- Search and filters make reports easier to navigate.
- Dashboards provide better visibility into recovery status.
- Can be expanded for real campus deployment.

17. LIMITATIONS

- Image quality can affect similarity results.
- Poor descriptions can reduce matching quality.

- Similar-looking objects can create false positives.
- Actual AI accuracy depends on the connected model/service and data.
- Production deployment requires stronger privacy, security and moderation controls.
- Advanced computer vision and real-time notifications may require external services.

18. FUTURE ENHANCEMENTS

- Dedicated computer-vision embedding model for robust image similarity.
- Campus map integration.
- QR-code based reporting points.
- Android and iOS applications.
- English, Telugu, Hindi and other regional language support.
- Secure identity verification before returning high-value items.
- Advanced analytics for frequently reported campus locations.

19. EXPECTED OUTCOME

The expected outcome is a functional, user-friendly web application that provides a centralized solution for campus lost-and-found management. The application demonstrates how AI-assisted matching can be incorporated into a practical student-focused web application and showcases application development, data management, authentication, workflows, and AI-assisted decision support.

20. CONCLUSION

Campus Lost & Found AI demonstrates the practical use of Artificial Intelligence in solving an everyday problem in an educational environment. By providing a centralized reporting platform and AI-assisted matching, the system can reduce the effort required to locate lost belongings and improve the chances of successful recovery.

The project also demonstrates the use of Base44 AI for rapid AI-assisted web application development. The application can serve as a foundation for a larger campus automation solution with advanced computer vision, mobile support, multilingual interaction, and secure identity verification.

21. REFERENCES

- Base44 AI – AI-assisted application development platform.
- Python Software Foundation – General Python programming reference, where applicable.
- General concepts of Artificial Intelligence, computer vision, image similarity, and recommendation/matching systems.
- Project Live Application: https://flat-campus-find-sync.base44.app
- GitHub – Software project hosting and version control platform.